SAIL · DURGAPUR STEEL PLANT · IRON MAKING CASE STUDY

# How Durgapur makes *iron*

*From ore and coal to hot metal: the four units of the iron zone, a blast furnace cut open, and what the planned BF-5 will change.*

Saichandan Mettu · MSME, IIT Hyderabad · Iron Making · November 2026

**2.40 Mt**
hot metal a year, rated (CAG gives 2.45)

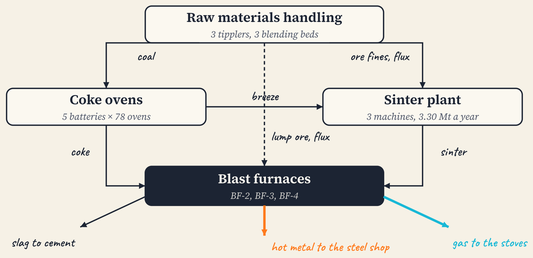
**3 furnaces**
BF-2, BF-3, BF-4: 4,600 m³ of useful volume

**6,856 t/day**
productivity × working volume, all three

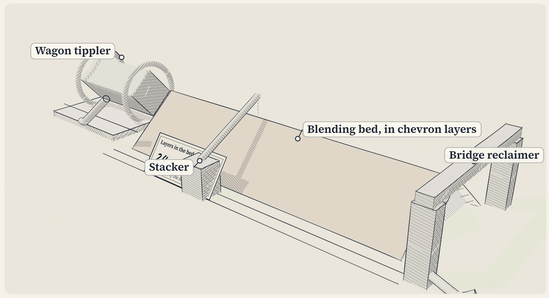
**105 %**
of its MoU target in 2017-24 (17.05 Mt), the best in SAIL

## 01 THE IRON ZONE

Durgapur Steel Plant, one of SAIL's five integrated plants, stands on the Damodar in West Bengal. Built with British help, it lit its first blast furnace on 29 December 1959. It makes its own coke and sinter and smelts them in three blast furnaces.

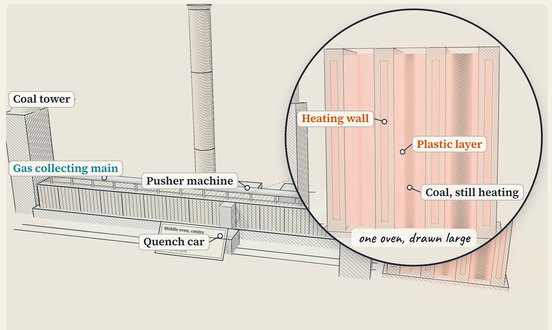



## 02 RAW MATERIALS



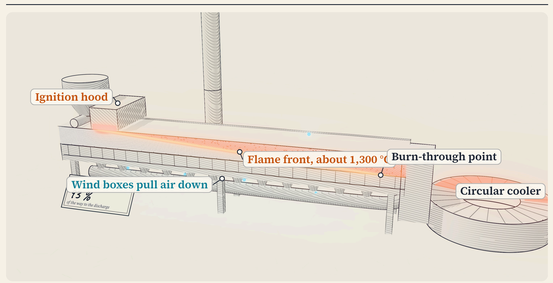

- Three wagon tipplers unload ore, flux and coal. Beds are laid in thin **chevron layers** and reclaimed across the face, so each slice averages every layer.
- Ore from SAIL's own mines (Bolani, Kiriburu, Gua); coking coal from the Jharia-Raniganj belt next door, blended with imported low-ash coal.

## 03 COKE OVENS



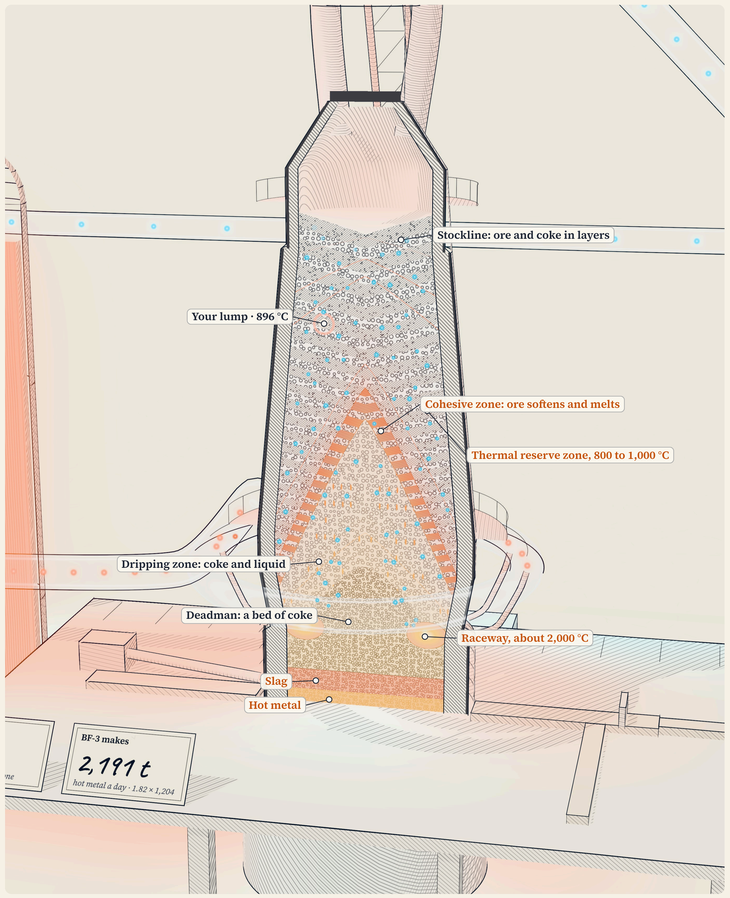

- **5 batteries × 78 ovens**, 4.45 m tall. Coal is heated without air to about 1,000 to 1,100 °C.
- Heat comes in from both walls: a **plastic layer** forms at each and they meet in the middle. Push too soon and the centre is still coal.
- Coke is fuel, reducing agent and the skeleton of the burden. The best furnaces use 300 to 350 kg per tonne of iron, Indian ones 450 to 550; SAIL averaged 419 in FY2025-26.

## 04 SINTER




- Three machines (142.7, 142.7 and 198 m²), 3.30 Mt a year: about **1.38 t of sinter per tonne of iron**.
- The **flame front**, about 1,300 °C, burns down while the bed rides along, and should reach the grate just before the end.

## 05 INSIDE THE BLAST FURNACE




BF-3 cut on its axis, drawn from its published volumes: **1,400 m³ useful** (taphole to stockline) and **1,204 m³ working** (tuyeres to stockline), so 196 m³ of hearth lies below the tuyeres.

Burden comes down in **6 to 8 hours**; the gas rises in **6 to 12 seconds**. The marked lump, about 2 hours in, is at **896 °C**, where CO turns wüstite into iron.

## 06 DURGAPUR'S FURNACES

| Furnace | Useful m³ | Working m³ | t/m³·day | t/day | Built, rebuilt |
|---|---|---|---|---|---|
| **BF-2** | 1,400 | 1,207 | 1.71 | 2,064 | 1961, 1993 |
| **BF-3** | 1,400 | 1,204 | 1.82 | 2,191 | 1962, 2002 |
| **BF-4** | 1,800 | 1,539 | 1.69 | 2,601 | 1967, 1996 |
| **All three** | **4,600** | **3,950** | | **6,856** | |

*Daily output = productivity × working volume; × 365 = 2.50 Mt a year. The best July, 2026: 2,27,400 t, or 7,335 t a day (7 % above). BF-1 (1959) is phased out.*

## 07 THE CHEMISTRY

**AT THE TUYERES**

$C + O_2 \rightarrow CO_2$ ; $CO_2 + C \rightarrow 2CO$

net $2C + O_2 \rightarrow 2CO$ *about 2,000 °C*

**INDIRECT REDUCTION, IN THE STACK**

$3Fe_2O_3 + CO \rightarrow 2Fe_3O_4 + CO_2$

$Fe_3O_4 + CO \rightarrow 3FeO + CO_2$

$FeO + CO \rightarrow Fe + CO_2$

**DIRECT REDUCTION, ABOVE ABOUT 1,000 °C**

$FeO + C \rightarrow Fe + CO$ *takes heat, so coke*

**FLUX AND SLAG**

$CaCO_3 \rightarrow CaO + CO_2$ ; $CaO + SiO_2 \rightarrow CaSiO_3$

*Hot metal (typical): 4.0 to 4.5 % C, 0.4 to 0.8 % Si, 1,450 to 1,500 °C. Slag: CaO/SiO₂ about 1.0 to 1.1, with 17 to 22 % Al₂O₃ from Indian ores.*

## WHAT IT TEACHES

1. **Location decides cost:** coal next door, ore from SAIL's own mines.
2. **Burden preparation drives output:** its own sinter and coke let three small furnaces beat their rating.
3. **Old furnaces run cool and soft:** 898 to 934 °C at 1.44 to 1.66 kg/cm², against 1,137 to 1,155 °C and about 4 in SAIL's new ones.
4. **Size wins:** one BF-5 will make about 94 % of what all three make today.

## PER TONNE OF HOT METAL

**1.38 t**
**sinter**
*3.30 Mt ÷ 2.40 Mt, rated*

**493 kg**
**lump ore**
*2023-24 (CAG)*

**≈ 350 kg**
**slag**
*2017-24 (CAG, our sum)*

**300–350**
**kg coke, world's best**
*India 450 to 550 (NITI)*

## IRONMAKING SWOT

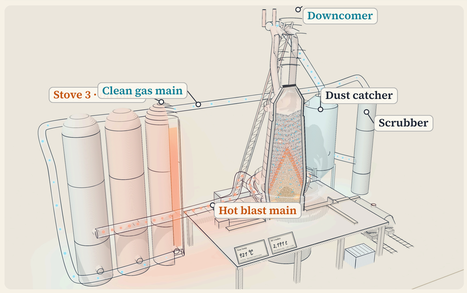
**STRENGTHS**
Captive ore; coal next door; its own coke and sinter; above its rating in 2022-24.

**WEAKNESSES**
Small furnaces from 1961-67; cool, low-pressure blast; one coke battery under repair.

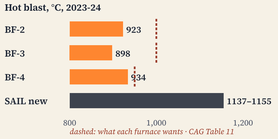
**OPPORTUNITIES**
BF-5 and a 2.5 Mt pellet plant; more sinter, hotter blast; demand for rails and TMT.

**THREATS**
Imported coal prices; expansion delays; tighter carbon rules; newer private plants.

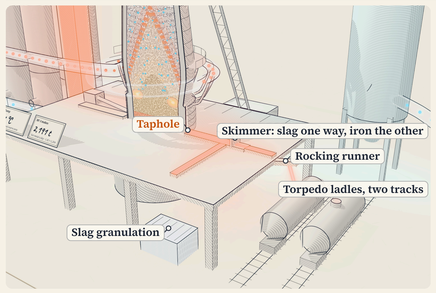
## 08 HOT BLAST AND THE GAS LOOP




The top gas still holds CO. Downcomer, dust catcher and scrubber clean it, and the **clean gas main** takes it back to the stoves, which store its heat in checker bricks and hand it to the **blast**.



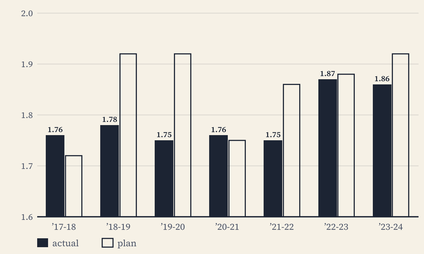

## 09 TAPPING AND SLAG




Twin tapholes on BF-3 (its 2002 rebuild): one casts while the other is plugged. A **skimmer** holds back the floating slag; a **rocking runner** fills two torpedo ladles in turn.

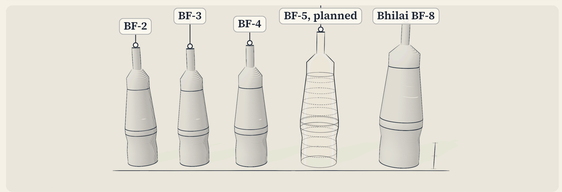
**≈ 350 kg** of slag per tonne of iron (5.93 Mt ÷ 17.05 Mt, 2017-24). Most is granulated for cement; 1.31 Mt was sold molten, which no other SAIL plant does.

## 10 PERFORMANCE




*Productivity, t per m³ of working volume a day, plan against actual (CAG Report 38 of 2025, Table 6). Output fell to 1.84 Mt in 2020 and reached 2.57 Mt in 2023.*

## 11 BF-5: ONE BIG FURNACE




To one scale: 1,400, 1,400 and 1,800 m³ today; BF-5 and Bhilai's BF-8, 3,000 and 4,060 m³. Planned for about 2030: **3,000 m³, 2.36 Mt a year** (≈ 6,466 t a day, 3 times BF-3 from 2.1 times the volume), with a 2.5 Mt pellet plant. BF-2 retires and BF-4 goes on standby: **3.42 Mt** of iron a year, +43 %.

**Sources.** SAIL, Durgapur plant pages. CAG Report No. 38 of 2025, performance audit of SAIL's blast furnaces, 2017-24 (furnace years, hot blast, pressure, productivity, slag). PIB, 9 Jul and 23 Oct 2002. Global Energy Monitor, Durgapur tracker and 2025 EIA summary (its March 2025 BF-3 reline is unconfirmed). Suchnaji (July 2026). SAIL Board's Report FY2025-26. NITI Aayog. J. A. Ricketts, The Blast Furnace; Techniques de l'Ingénieur M7403 (burden and gas times). Zone temperatures and analyses are typical textbook ranges. Capacities differ by source (CAG: 2.45 Mt hot metal, 3.01 Mt sinter, 1.66 Mt coke).

Figures rendered from the 3D model built for this poster, in which every part stands up off the paper and the furnace can be cut open. Furnace shapes are typical profiles scaled to Durgapur's published volumes; units other than BF-3 are drawn shortened.